

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q001 Written

Given the probability distributions of X and Y in the given tables, find $E(X+Y)$.

Solution

1. Find EX and EY separately, then add them.

Final answer

$$EX = 35/3, EY = 9/2, \text{ so } E(X + Y) = 97/6.$$

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

Solution

1. Each fair head indicator has expected value $1/2$.

Final answer

$$E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q003 Written

For the two given tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Solution

1. The expectation of a sum is the sum of the expectations.

Final answer

$$E(X + Y) = EX + EY = 2 + 5 = 7.$$

Q004 Written

For the two bags in the given exercise, find the expected total number of red balls drawn.

Solution

1. Use $E(\text{red drawn}) = \text{sample size} \times \text{the red proportion in each bag}$.

Final answer

Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Solution

1. Add the three binomial means.

Final answer

Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

Solution

1. Model the score as $aX+bY$, then apply linearity.

Final answer

$Z = 500X + 100Y$, so $EZ = 5001 + 1003/2 = 650$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q007 Written

Two 500-point coins and two 100-point coins are tossed. Find EZ .

Solution

1. The expected heads in each two-coin group is 1.

Final answer

$$EZ = 500(1) + 100(1) = 600.$$

Q008 Written

A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .

Solution

1. Both die rolls have mean $7/2$.

Final answer

$$EZ = 3(7/2) + 5(7/2) = 28.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Solution

1. Use the expected score of each fair coin and add.

Final answer

$$EZ = 350 + 25 = 175.$$

Q010 Written

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

Solution

1. Expected winners are sample size times the winning proportion in each raffle.

Final answer

$$EZ = 22/10 + 21/6 = 11/15.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Solution

1. Check independence before multiplying expectations.

Final answer

X and Y are independent, so $E(XY) = EXEY = 3/2 \times 7/2 = 21/4$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at 1,2,3 and Y uniform on 2,4,6, find $E(XY)$.

Solution

1. Compute each mean from its table, then multiply.

Final answer

$EX = 8/5, EY = 4$, so $E(XY) = 32/5$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Solution

1. The two experiments are independent.

Final answer

$$E(XY) = EXEY = 3/2 \cdot 1 = 3/2.$$

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

Solution

1. Replacement makes the two draws independent.

Final answer

$$EX = EY = 5/2, \text{ so } E(XY) = 25/4.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Solution

1. Use the product rule after checking independence.

Final answer

$$E(XY) = EXEY = 2/7.$$

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. For independent variables, add variances, then take the positive square root.

Final answer

$$VX=2/3, VY=3/4, \text{ so } V(X + Y) = 17/12 \text{ and } \sigma=\sqrt{17/12}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q017 Written

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Solution

1. Each fair die has variance $35/12$.

Final answer

$$V(X + Y) = 35/12 + 35/12 = 35/6; \sigma = \sqrt{35/6}.$$

Q018 Written

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Use $V(\text{Bernoulli}(1/2)) = 1/4$ and $V(\text{die}) = 35/12$.

Final answer

$$V = 1/4 + 35/12 = 19/6; \sigma = \sqrt{19/6}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Solution

1. Variances of independent Bernoulli counts add.

Final answer

$$V = 2/4 + 3/4 = 5/4.$$

Q020 Written

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Use the uniform discrete variance for each scale and add.

Final answer

$$V = 5/4 + 35/12 = 25/6; \sigma = 5/\sqrt{6}.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. For a product multiply means; for a sum add variances.

Final answer

$EX=4$, $VX=8/3$; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 Written

Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. All three distributions are identical and independent.

Final answer

$EX=3$, $VX=8/3$; $E(XYZ) = 27$ and $V(sum) = 8$.

Worked Solutions

Expected Value and Variance of Multiple Variables

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Solution

1. Use independence for the product and variance addition for the sum.

Final answer

$$E(\text{product}) = 7/2^3 = 343/8; V(\text{sum}) = 335/12 = 35/4.$$

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

Solution

1. For Binomial($n, 1/6$), use $E = np$ and $V = np(1 - p)$.

Final answer

$$\text{Means are } 5/6, 2/3, 1/2; E(\text{product}) = 5/18 \text{ and } V(\text{sum}) = 5/3.$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q025 **Written**

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Solution

1. Multiply the three expected head counts.

Final answer

$$E(XYZ) = 23/21=3.$$

Q026 **MCQ**

$E(X+Y)$ equals:

Solution

1. Add the two expectations.

Correct option: B

Final answer

$$EX + EY$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Solution**

1. Extend linearity to three terms.

Correct option: B

Final answer

$$EX + EY + EZ$$

Q028 MCQ

The expected heads in one fair coin toss is:**Solution**

1. Use the Bernoulli mean.

Correct option: B

Final answer

$$1/2$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Solution**1. Use np .**Correct option:** B**Final answer**

1

Q030 MCQ

If $EX=2$ and $EY=5$, $E(X+Y)$ is:**Solution**

1. Add the means.

Correct option: B**Final answer**

7

Worked Solutions

Expected Value and Variance of Multiple Variables

Q031 MCQ

 $E(aX+bY)$ equals:**Solution**

1. Apply linearity.

Correct option: A

Final answer

$$aEX + bEY$$

Q032 MCQ

The expected score of one fair 100-point coin is:**Solution**

1. Half the value is expected.

Correct option: B

Final answer

50

Worked Solutions

Expected Value and Variance of Multiple Variables

Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Solution**

1. Use both means $7/2$.

Correct option: C**Final answer**

28

Q034 MCQ

A fixed entry fee changes:**Solution**

1. It is a constant term.

Correct option: B**Final answer**

the mean only

Worked Solutions

Expected Value and Variance of Multiple Variables

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Solution

1. Use sample size times winner proportion.

Correct option: B

Final answer

2/5

Q036 MCQ

$E(XY) = E(X)E(Y)$ requires:

Solution

1. The product rule needs mutual independence.

Correct option: B

Final answer

independence

Worked Solutions

Expected Value and Variance of Multiple Variables

Q037 MCQ

For three fair coins, EX is:**Solution**

1. Use np.

Correct option: C**Final answer** $3/2$

Q038 MCQ

For a fair die, EY is:**Solution**

1. Average 1 through 6.

Correct option: C**Final answer**

3.5

Worked Solutions

Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Solution**

1. Replacement restores the same distribution and independence.

Correct option: B

Final answer

independent

Q040 MCQ

If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:**Solution**

1. Multiply the means.

Correct option: B

Final answer

6

Worked Solutions

Expected Value and Variance of Multiple Variables

Q041 MCQ

For independent X, Y , $V(X+Y)$ equals:**Solution**

1. Add independent variances.

Correct option: B**Final answer**

$$VX + VY$$

Q042 MCQ

The variance of a fair die is:**Solution**

1. Use the standard die variance.

Correct option: B**Final answer**

$$35/12$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Solution**

1. Take the positive root.

Correct option: B

Final answer $\sqrt{V(X + Y)}$

Q044 MCQ

The variance of the sum of two independent fair dice is:**Solution**

1. Double 35/12.

Correct option: B

Final answer $35/6$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Solution**

1. Translations do not change spread.

Correct option: C**Final answer**

neither spread measure

Q046 MCQ

For independent X, Y, Z , $E(XYZ)$ equals:**Solution**

1. Multiply the three means.

Correct option: B**Final answer** $E(X)E(Y)E(Z)$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Solution**

1. Add the three variances.

Correct option: B

Final answer

$$VX + VY + VZ$$

Q048 MCQ

For a binomial count with $n=5$ and $p=1/6$, EX is:**Solution**1. Use np .

Correct option: B

Final answer

$$5/6$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Solution**

1. Average the three values.

Correct option: C**Final answer**

4

Q050 MCQ

For three independent fair dice, $V(\text{sum})$ is:**Solution**

1. Add three copies of $35/12$.

Correct option: C**Final answer** $35/4$

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 Concept Quiz · Written

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Solution

1. Use linearity: 34-27.

Final answer

−2

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

Solution

1. Multiply the means.

Final answer

10

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Solution

1. Add the two die variances.

Final answer

$$V = 35/6, \sigma = \sqrt{35/6}$$

Quiz WRITTEN 4 **Concept Quiz · Written**

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

Solution

1. Add $np(1-p)$ for the three counts.

Final answer

$$9/4$$

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz · MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Solution

1. Compute $2 \times 3 + 4$.

Correct option: B

Final answer

10

Quiz MCQ 6 [Concept Quiz · MCQ](#)

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Solution

1. Multiply the means.

Correct option: B

Final answer

10

Worked Solutions

Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz · MCQ](#)

For two independent die rolls, $V(X+Y)$ is:

Solution

1. Add the two variances.

Correct option: B

Final answer

35/6

Quiz MCQ 8 [Concept Quiz · MCQ](#)

For three independent variables, $V(X+Y+Z)$ is:

Solution

1. Extend variance addition.

Correct option: B

Final answer

the sum of variances

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, \text{ so } E(X + Y) = 97/6.$ Q016 $VX=2/3, VY=3/4, \text{ so } V(X + Y) = 17/12.$ Q025 $E(XYZ) = 23/21=3.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$ and $\sigma = \sqrt{17/12}.$ Q026 B

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$ Q017 $V(X + Y) = 35/12 + 35/12 = 35/6;$ Q027 B

Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$ $\sigma = \sqrt{35/6}.$ Q028 B

Q005 Each mean is $2(0.5) = 1, \text{ so } E(X + Y + Z) = 3.$ Q018 $V = 1/4 + 35/12 = 19/6;$ Q029 B
 $\sigma = \sqrt{19/6}.$ Q030 B

Q006 $Z = 500X + 100Y, \text{ so } EZ = 5001 + 1003/2 = 6001.$ Q019 $V = 21/4 + 31/4 = 5/4.$ Q031 A

Q007 $EZ = 5001 + 1001 = 6001.$ Q020 $V = 5/4 + 35/12 = 25/6;$ Q032 B

Q008 $EZ = 37/2 + 57/2 = 28.$ $\sigma = 5/\sqrt{6}.$ Q033 C

Q009 $EZ = 350 + 25 = 175.$ Q021 $EX=4, VX=8/3; E(XYZ) = 64 \text{ and } V(sum) = 8.$ Q034 B

Q010 $EZ = 22/10 + 21/6 = 11/5.$ Q022 $EX=3, VX=8/3; E(XYZ) = 27 \text{ and } V(sum) = 8.$ Q035 B

Q011 X and Y are independent, so $E(XY) = EXEY = 3/27 \cdot 2 = 21/4.$ Q023 $E(product) = 7/2^3 = 343/8;$ Q036 B
 $V(sum) = 335/12 = 35/4.$ Q037 C

Q012 $EX = 8/5, EY = 4, \text{ so } E(XY) = 32/5.$ Q024 Means are $5/6, 2/3, 1/2;$ Q038 C
 $E(product) = 5/18 \text{ and } V(sum) = 5/3.$ Q039 B

Q013 $E(XY) = EXEY = 3/21 = 3/2.$ Q040 B

Q014 $EX = EY = 5/2, \text{ so } E(XY) = 25/4.$ Q041 B

Q015 $E(XY) = EXEY = 2/7.$

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$ Quiz WRITTEN 4 $9/4$

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B